

OPERATIONS RESEARCH • TRANSPORTATION SYSTEMS

Optimal Driver–Vehicle Allocation System for Public Transportation Using Assignment Models

A Linear Assignment-Model Approach to Cost-Efficient, Workload-Balanced Fleet Scheduling

Presented to the Department Review Committee
Academic Year 2025–2026



Driver Scheduling • Fleet Optimization • Hungarian Algorithm

Why Manual Driver–Vehicle Assignment Falls Short



Uneven Workload

Some drivers consistently log far more distance and hours than others.



Vehicle Underutilization

Fleet vehicles sit idle or under-used while others are overworked.



Higher Operating Cost

Ad-hoc pairing ignores cost-minimizing combinations of driver and vehicle.



Driver Fatigue

Excessive working hours on certain shifts raise safety and attrition risk.



Inefficient Fleet Management

No systematic, data-driven basis exists for who drives what.



Research Objective

Develop a mathematical model that assigns drivers to vehicles optimally — minimizing operational cost while balancing workload across the fleet.

Six Steps from Raw Shift Data to an Optimized Schedule

01



Analyze historical driver and vehicle shift data

02



Develop a cost matrix from operational parameters

03



Formulate the driver–vehicle assignment problem

04



Obtain optimal assignments via the Hungarian Algorithm

05



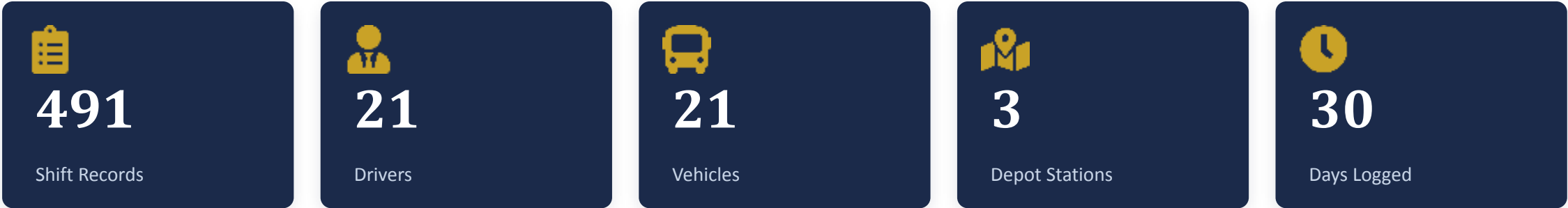
Compare existing schedules against optimized schedules

06



Evaluate workload balance and efficiency gains

The Operational Dataset Behind This Study



Distance per Shift (km)		Shift Duration (hours)	
Mean	117.4 km	Mean	10.6
Std. Dev.	27.9 km	Std. Dev.	1.9
Minimum	10.3 km	Median	10.5
Maximum	191.9 km	Stations Covered	504 · 511 · 503

Source: ShiftData.csv — one month of pilot, vehicle, and station-level shift logs

Variables Captured in the Shift Log

Variable	Field Name	Role in the Model
Driver ID	Pilotcode	Index i — identifies each of the 21 drivers
Vehicle ID	VehicleCode	Index j — identifies each of the 21 vehicles
Depot / Station	StationCode	Groups assignments by operating depot
Shift Start	StartTime	Used to compute shift duration T_i
Shift End	EndTime	Used to compute shift duration T_i
Distance Travelled	Distance	Used to compute average distance D_i
Revenue / Trips	Total	Used in cost-efficiency evaluation

These raw fields are aggregated per driver and per vehicle to build the cost matrix used by the assignment model.

Formulating the Assignment Problem

Decision Variable

$$x_{ij} = 1 \quad \text{if driver } i \text{ is assigned to vehicle } j$$

$$x_{ij} = 0 \quad \text{otherwise}$$

Binary condition: $x_{ij} \in \{0, 1\}$

Cost Matrix

$$C = (c_{ij}), \quad c_{ij} = w_1 D_i + w_2 T_i + w_3 R_i$$

D_i = average distance travelled

T_i = average shift duration

R_i = vehicle utilization factor

$$w_1 + w_2 + w_3 = 1$$

Objective Function

Minimize Z

$$Z = \sum_{i=1..m} \sum_{j=1..n} c_{ij} \cdot x_{ij}$$

Total assignment cost summed across all driver–vehicle pairs

Constraints



Each driver gets exactly one vehicle:

$$\sum_{j=1..n} x_{ij} = 1 \quad \text{for } i = 1, 2, \dots, m$$



Each vehicle gets exactly one driver:

$$\sum_{i=1..m} x_{ij} = 1 \quad \text{for } j = 1, 2, \dots, n$$

Solving the Assignment with the Hungarian Algorithm

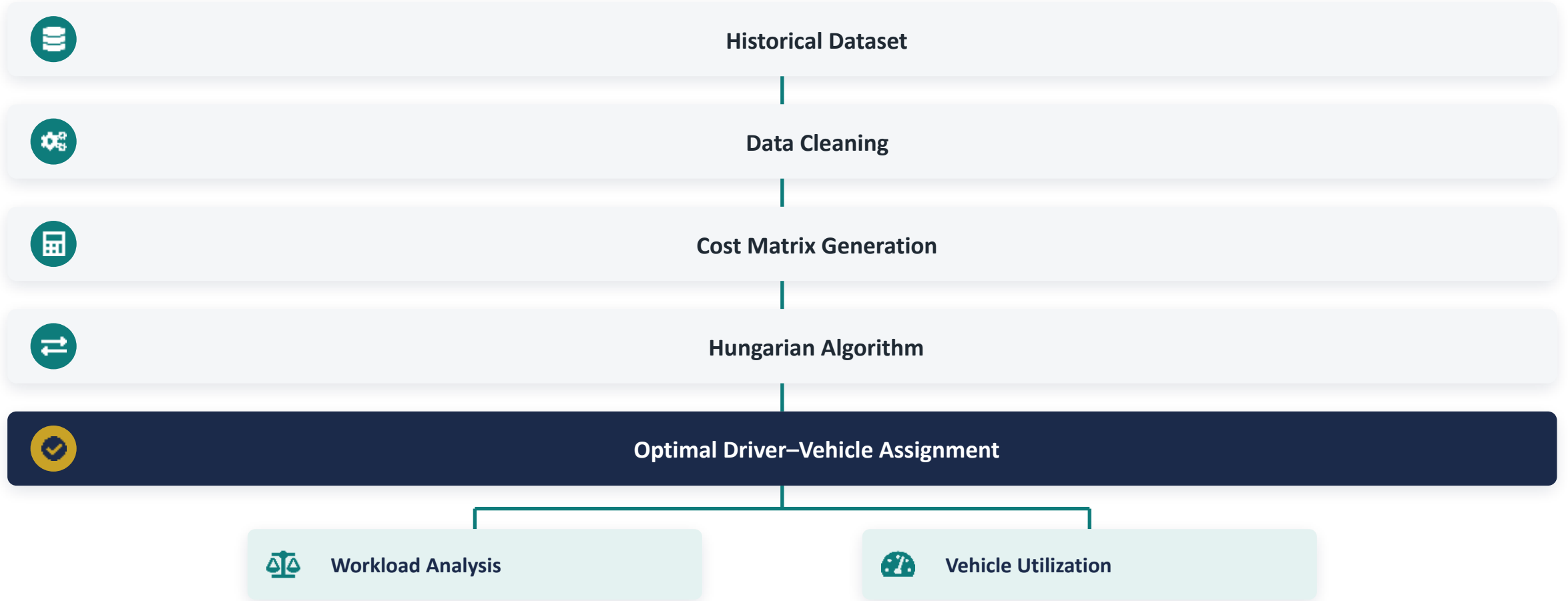
Illustrative Cost Matrix C

	V1	V2	V3
D1	15	18	20
D2	10	14	16
D3	12	17	13

D_i, T_i and R_i values are combined into each cell per the cost formula

-  **Construct the Cost Matrix**
-  **Row Reduction**
-  **Column Reduction**
-  **Draw Minimum Covering Lines**
-  **Obtain Optimal Assignment**

Proposed System Architecture



Measuring and Minimizing Workload Imbalance

Shift Duration

$$SD = \text{End Time} - \text{Start Time}$$

Average Workload

$$W_i = \text{Total Distance}_i / \text{Number of Shifts}$$

Workload Variance

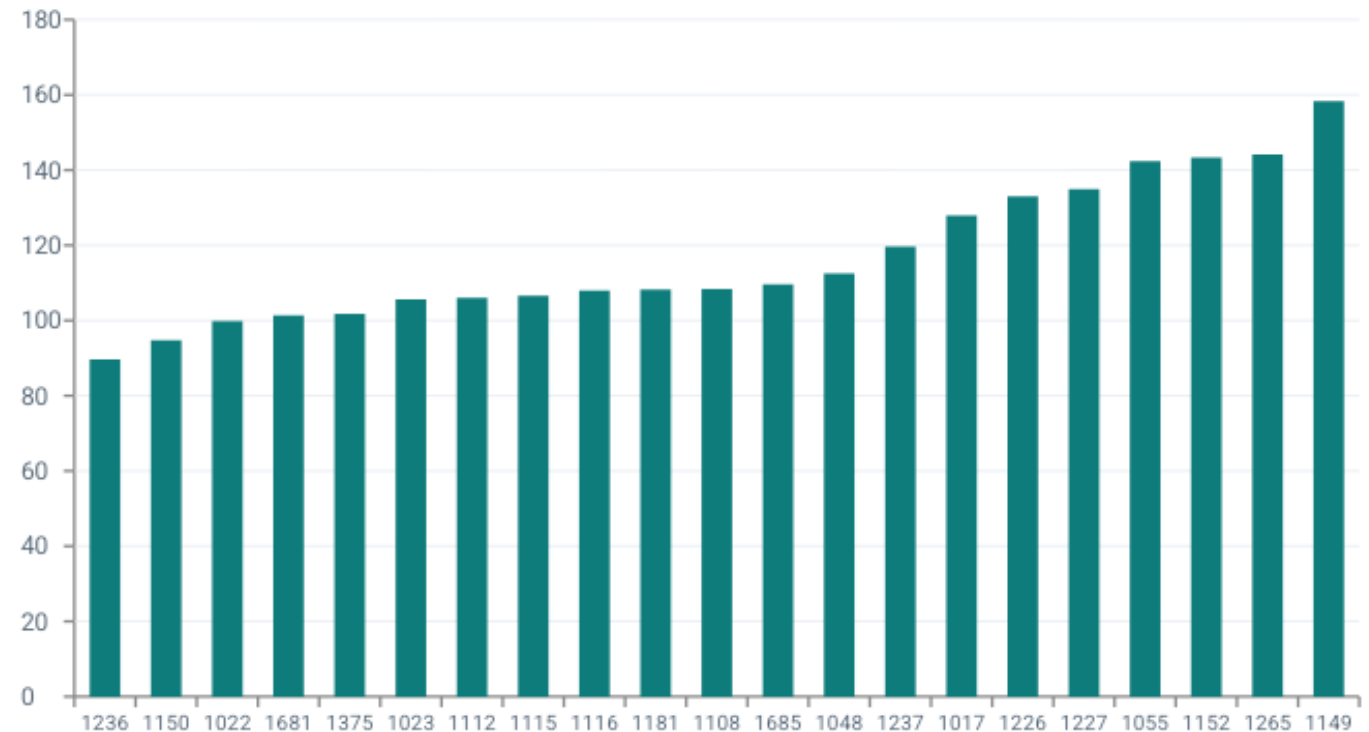
$$\sigma^2 = (1/n) \sum_{i=1..n} (W_i - \bar{W})^2$$

Objective: $\min \sigma^2$ — smaller variance $\Rightarrow$ more balanced workload

Workload Balance Index (WBI)

0.84 (computed from current shift logs)

Average Distance per Shift, by Driver (km)



Drivers ordered by average shift distance — the spread illustrates current workload imbalance the model aims to reduce.

Classifying Fleet Utilization

Utilization Measure

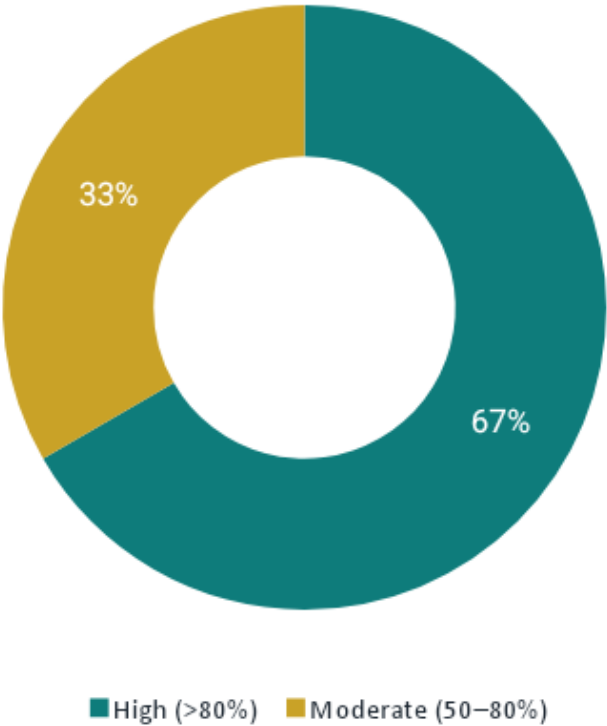
$$U_j = \text{Distance Travelled} / \text{Available Time}$$

Computed for each vehicle j across all logged shifts in the dataset

Utilization	Category
>80%	High
50% – 80%	Moderate
<50%	Low

Finding: across the 21 fleet vehicles, 14 fall in the High band and 7 in Moderate — no vehicle in this dataset is Low-utilized.

Fleet Utilization Split (21 Vehicles)



How Success Will Be Measured



Cost Reduction (CR)

$$CR = [(OC - PC) / OC] \times 100$$

OC = existing cost, PC = proposed cost. Higher CR means greater savings from optimized allocation.



Workload Balance Index (WBI)

$$WBI = 1 - (\sigma / \bar{W})$$

Ratio of workload standard deviation to mean. Higher WBI indicates more even workload distribution.



Vehicle Utilization Improvement (VUI)

$$VUI = [(U_{new} - U_{old}) / U_{old}] \times 100$$

Percentage gain in vehicle utilization after applying the optimized assignment.

Software Tools for Implementation



Python

Core scripting & data processing



Excel

Data exploration & reporting



MATLAB

Matrix computation & validation



OR-Tools

Assignment & optimization solver



PuLP

Linear programming formulation



SciPy

linear_sum_assignment (Hungarian Algorithm)

What the Optimized Model Delivers



Optimal Driver–Vehicle Allocation

Every driver matched to the cost-minimizing vehicle under the assignment constraints.



Reduction in Operational Cost

Lower combined cost relative to the existing manual assignment baseline.



Balanced Driver Workload

Reduced variance in distance and hours across the driver pool.



Better Vehicle Utilization

Fewer underused vehicles; utilization shifted toward the High band.



A Decision-Support System

A reusable mathematical framework for ongoing transport management.

A Publication-Level Research Extension



“Neutrosophic Driver–Vehicle Allocation Model under Uncertain Traffic and Driver Availability Conditions”

What this extension changes

- Assignment costs represented as interval neutrosophic numbers, not crisp values
- Explicitly models uncertainty in traffic conditions and driver availability
- Strengthens the mathematical rigor for journal-level publication



Thank You

Questions & Discussion

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